arXiv:2210.16156v2 [cs.LG] 16 Nov 2022

RELIABILITY OF CKA AS A SIMILARITY MEASURE IN DEEP LEARNING

MohammadReza Davari [1,3] * **Stefan Horoi** [2,3] * **Amine Natik** [2,3]
Guillaume Lajoie [2,3] **Guy Wolf** [2,3] † **Eugene Belilovsky** [1,3] †

[1] Concordia University [2] Université de Montréal [3] Mila – Quebec AI Institute
{mohammadreza.davari, eugene.belilovsky}@concordia.ca
{stefan.horoi, amine.natik, guillaume.lajoie, guy.wolf}@umontreal.ca

ABSTRACT

Comparing learned neural representations in neural networks is a challenging but important problem, which has been approached in different ways. The Centered Kernel Alignment (CKA) similarity metric, particularly its linear variant, has recently become a popular approach and has been widely used to compare representations of a network's different layers, of architecturally similar networks trained differently, or of models with different architectures trained on the same data. A wide variety of conclusions about similarity and dissimilarity of these various representations have been made using CKA. In this work we present analysis that formally characterizes CKA sensitivity to a large class of simple transformations, which can naturally occur in the context of modern machine learning. This provides a concrete explanation of CKA sensitivity to outliers, which has been observed in past works, and to transformations that preserve the linear separability of the data, an important generalization attribute. We empirically investigate several weaknesses of the CKA similarity metric, demonstrating situations in which it gives unexpected or counter-intuitive results. Finally we study approaches for modifying representations to maintain functional behaviour while changing the CKA value. Our results illustrate that, in many cases, the CKA value can be easily manipulated without substantial changes to the functional behaviour of the models, and call for caution when leveraging activation alignment metrics.

1 INTRODUCTION

In the last decade, increasingly complex deep learning models have dominated machine learning and have helped us solve, with remarkable accuracy, a multitude of tasks across a wide array of domains. Due to the size and flexibility of these models it has been challenging to study and understand exactly *how* they solve the tasks we use them on. A helpful framework for thinking about these models is that of *representation learning*, where we view artificial neural networks (ANNs) as learning increasingly complex internal representations as we go deeper through their layers. In practice, it is often of interest to analyze and compare the representations of multiple ANNs. However, the typical high dimensionality of ANN internal representation spaces makes this a fundamentally difficult task.

To address this problem, the machine learning community has tried finding meaningful ways to compare ANN internal representations and various *representation (dis)similarity measures* have been proposed (Li et al., 2015; Wang et al., 2018; Raghu et al., 2017; Morcos et al., 2018). Recently, Centered Kernel Alignment (CKA) (Kornblith et al., 2019) was proposed and shown to be able to reliably identify correspondences between representations in architecturally similar networks trained on the same dataset but from different initializations, unlike past methods such as linear regression or CCA based methods (Raghu et al., 2017; Morcos et al., 2018). While CKA can capture different notions of similarity between points in representation space by using different kernel functions, it was empirically shown in the original work that there are no real benefits to using CKA with a nonlinear kernel over its linear counterpart. As a result, linear CKA has been the preferred representation similarity measure of the machine learning community in recent years and other similarity measures (including nonlinear CKA) are seldomly used. CKA has been utilized in a number of works to make conclusions regarding the similarity between different models and their behaviours such as wide versus deep ANNs (Nguyen et al., 2021) and transformer versus CNN based ANNs (Raghu et al.,

*Equal contribution, name order randomized. †Equal senior-author contribution

2021). They have also been used to draw conclusions about transfer learning (Neyshabur et al., 2020) and catastrophic forgetting (Ramasesh et al., 2021). Due to this widespread use, it is important to understand how reliable the CKA similarity measure is and in what cases it fails to provide meaningful results. In this paper, we study CKA sensitivity to a class of simple transformations and show how CKA similarity values can be directly manipulated without noticeable changes in the model final output behaviour. In particular our contributions are as follows:

In Sec. 3 and with Thm. 1 we characterize CKA sensitivity to a large class of simple transformations, which can naturally occur in ANNs. With Cor. 3 and 4 we extend our theoretical results to cover CKA sensitivity to outliers, which has been empirically observed in previous work (Nguyen et al., 2021; Ding et al., 2021; Nguyen et al., 2022), and to transformations preserving linear separability of data, an important characteristic for generalization. Concretely, our theoretical contributions show how the CKA value between two copies of the same set of representations can be significantly decreased through simple, functionality preserving transformations of one of the two copies. In Sec. 4 we empirically analyze CKA's reliability, illustrating our theoretical results and subsequently presenting a general optimization procedure that allows the CKA value to be heavily manipulated to be either high or low without significant changes to the functional behaviour of the underlying ANNs. We use this to revisit previous findings (Nguyen et al., 2021; Kornblith et al., 2019).

2 Background on CKA and Related Work

Comparing representations Let $X \in \mathbb{R}^{n \times d_1}$ denote a set of ANN internal representations, i.e., the neural activations of a specific layer with d_1 neurons in a network, in response to $n \in \mathbb{N}$ input examples. Let $Y \in \mathbb{R}^{n \times d_2}$ be another set of such representations generated by the same input examples but possibly at a different layer of the same, or different, deep learning model. It is standard practice to center these representations column-wise (feature or "neuron" wise) before analyzing them. We are interested in representation similarity measures, which try to capture a certain notion of similarity between X and Y.

Quantifying similarity Li et al. (2015) have considered one-to-one, many-to-one and many-to-many mappings between neurons from different neural networks, found through activation correlation maximization. Wang et al. (2018) extended that work by providing a rigorous theory of neuron activation subspace match and algorithms to compute such matches between neurons. Alternatively, Raghu et al. (2017) introduced SVCCA where singular value decomposition is used to identify the most important directions in activation space. Canonical correlation analysis (CCA) is then applied to find maximally correlated singular vectors from the two sets of representations and the mean of the correlation coefficients is used as a similarity measure. In order to give less importance to directions corresponding to noise, Morcos et al. (2018) introduced projection weighted CCA (PWCCA). The PWCCA similarity measure corresponds to the weighted sum of the correlation coefficients, assigning more importance to directions in representation space contributing more to the output of the layer. Many other representation similarity measures have been proposed based on linear classifying probes (Alain & Bengio, 2016; Davari et al., 2022), fixed points topology of internal dynamics in recurrent neural networks (Sussillo & Barak, 2013; Maheswaranathan et al., 2019), solving the orthogonal Procrustes problem between sets of representations (Ding et al., 2021; Williams et al., 2021) and many more (Laakso & Cottrell, 2000; Lenc & Vedaldi, 2018; Arora et al., 2017). We also note that a large body of neuroscience research has focused on comparing neural activation patterns in biological neural networks (Edelman, 1998; Kriegeskorte et al., 2008; Williams et al., 2021; Low et al., 2021).

CKA Centered Kernel Alignment (CKA) (Kornblith et al., 2019) is another such similarity measure based on the Hilbert-Schmidt Independence Criterion (HSIC) (Gretton et al., 2005) that was presented as a means to evaluate independence between random variables in a non-parametric way. For $K_{i,j} = k(x_i, x_j)$ and $L_{i,j} = l(y_i, y_j)$ where k, l are kernels and for $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ the centering matrix, HSIC can be written as: $\text{HSIC}(K, L) = \frac{1}{(n-1)^2} tr(KHLH)$. CKA can then be computed as:

$$\text{CKA}(K, L) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}(K, K)\text{HSIC}(L, L)}} \tag{1}$$

In the linear case k and l are both the inner product so $K = XX^\top$, $L = YY^\top$ and we use the notation $CKA(X, Y) = CKA(XX^\top, YY^\top)$. Intuitively, HSIC computes the similarity structures of X and Y, as measured by the kernel matrices K and L, and then compares these similarity structures (after centering) by computing their alignment through the trace of $KHLH$.

Recent CKA results CKA has been used in recent years to make many claims about neural network representations. Nguyen et al. (2021) used CKA to establish that parameter initialization drastically impact feature similarity and that the last layers of overparameterized (very wide or deep) models learn representations that are very similar, characterized by a visible "block structure" in the networks CKA heatmap. CKA has also been used to compare vision transformers with convolutional neural networks and to find striking differences between the representations learned by the two architectures, such as vision transformers having more uniform representations across all layers (Raghu et al., 2021). Ramasesh et al. (2021) have used CKA to find that deeper layers are especially responsible for forgetting in transfer learning settings.

Most closely related to our work, Ding et al. (2021) demonstrated that CKA lacks *sensitivity* to the removal of low variance principal components from the analyzed representations even when this removal significantly decreases probing accuracy. Also, Nguyen et al. (2022) found that the previously observed high CKA similarity between representations of later layers in large capacity models (so-called block structure) is actually caused by a few dominant data points that share similar characteristics. Williams et al. (2021) discussed how CKA does not respect the triangle inequality, which makes it problematic to use CKA values as a similarity measure in downstream analysis tasks. We distinguish ourselves from these papers by providing theoretical justifications to CKA sensitivity to outliers and to directions of high variance which were only empirically observed in Ding et al. (2021); Nguyen et al. (2021). Secondly, we do not only present situations in which CKA gives unexpected results but we also show *how* CKA values can be manipulated to take on arbitrary values.

Nonlinear CKA The original CKA paper (Kornblith et al., 2019) stated that, in practice, CKA with a nonlinear kernel gave similar results as linear CKA across the considered experiments. Potentially as a result of this, all subsequent papers which used CKA as a neural representation similarity measure have used linear CKA (Maheswaranathan et al., 2019; Neyshabur et al., 2020; Nguyen et al., 2021; Raghu et al., 2021; Ramasesh et al., 2021; Ding et al., 2021; Williams et al., 2021; Kornblith et al., 2021), and to our knowledge, no published work besides Kornblith et al. (2019); Nguyen et al. (2022) has used CKA with a nonlinear kernel. Consequently, we largely focus our analysis on linear CKA which is the most popular method and the one actually used in practice. However, our empirical results suggest that many of the observed problems hold for CKA with an RBF kernel and we discuss a possible way of extending our theoretical results to the nonlinear case.

3 CKA SENSITIVITY TO SUBSET TRANSLATION

Invariances and sensitivities are important theoretical characteristics of representation similarity measures since they indicate what kind of transformations respectively maintain or change the value of the similarity measure. In other words they indicate what transformations "preserve" or "destroy" the similarity between representations as defined by the specific measure used. Kornblith et al. (2019) argued that representation similarity measures should not be invariant to invertible linear transformations. They showed that measures which are invariant to such transformations give the same result for any set of representations of width (i.e. number of dimensions/neurons) greater than or equal to the dataset size. They instead introduced the CKA method, which satisfies weaker conditions of invariance, specifically it is invariant to orthogonal transformations such as permutations, rotations and reflections and to isotropic scaling. Alternatively, transformations to which representation similarity measures are sensitive have not been studied despite being highly informative as to what notion of "similarity" is captured by a given measure. For example, a measure that is highly sensitive to a transformation $\mathcal{T}$ is clearly measuring a notion of similarity that is destroyed by $\mathcal{T}$. In this section, we theoretically characterize CKA sensitivity to a wide class of simple transformations, namely the translation of a subset of the representations. We also justify why this class of transformations and the special cases it contains are important in the context of predictive tasks that are solved using neural networks.

Theorem 1. *Consider a set of n internal representations in p dimensions $X \in \mathbb{R}^{n \times p}$ that have been centered column-wise, let $S \subset X$ such that $\rho = \frac{|S|}{|X|} \leq \frac{1}{2}$ and $\vec{v}$ such that $\|\vec{v}\| = 1$. We define $X_{S,\vec{v},c} = S \cup \{x + c\vec{v} : x \in X \backslash S\}$. Then we have:*

$$\lim_{c \to \infty} CKA_{lin}(X, X_{S,\vec{v},c}) = \Gamma(\rho) \frac{\|\mathbb{E}_{x \in S}[x]\|^2}{\mathbb{E}_{x \in X}[\|x\|^2]} \sqrt{\dim_{PR}(X)} \tag{2}$$

where $\Gamma(\rho) = \frac{\rho}{1-\rho} \in (0, 1]$, and $\dim_{PR}(X) \triangleq \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \in [1, p]$ the dimensionality estimate provided by the participation ratio of eigenvalues $\{\lambda_i\}$ of the covariance of X.